Hongli Ye

Shenzhen, GuangDong 12311501@mail.sustech.edu.cn 139 5774 7881 Personal Webpage

Educational Background

BSc **Southern University of Science and Technology**, Mathematics — Guangdong, China, Sep. 2023 - July 2027 (Expected)

- GPA: 3.74/4.0 or 89.14/100.
- English Proficiency: **Duolingo**: 130/160; **CET6**: 606/710; **IELTS**: 6.5/9.0
- Relevant Coursework: Abstract Algebra(H)(90), Complex Analysis(97), Differential Manifold(92), Riemannian Geometry(96), Algebraic Topology(91)

University of Zurich, Mathematics — Zurich, Switzerland, Sep. 2026 - Jan. 2027

- Incoming Regular Visiting Student
- Ongoing Courses: Algebraic Topology, Category Theory, General Relativity, Knot Theory(ETHz)

North Carolina State University, Mathematics — North Carolina, USA, Jan. 2025 - Feb. 2025

- Research Topic: Reinforcement Learning, Cart-Pole system (**Supervisor**: Dr. Tran)
- Participating in coding and training a model using reinforcement learning that can automatically balance the cart-pole system by controlling the velocity of the cart. Gaining the basic idea of how the Rie

Member of the Honors Program, Department of Mathematics — Guangdong, China, Nov. 2023 - Feb. 2025

- Enrolling in Advanced Mathematics Courses.

Research Interests

Research interests include geometric topology and differential geometry. But I am still eager to explore my potential areas of interest.

Teaching Experience

Teaching Assistant, MA202 Complex Analysis — Guangdong, China, Spring 2026

- Assisted in grading homework assignments and examinations for undergraduate-level Complex Analysis.
- Prepared reference solutions, graded homework and exams, and supported course instructions for students.

Seminar Participation

Riemann Surfaces Seminar Organizer: Yifei Zhu — Guangdong, China, Feb. 2026 - May. 2026

- Organize the seminar and get the experience of presenting lectures
- Reading sections 1-10 and 11-17 of the book *Riemann Surface written by Forster*, getting the basic idea of the Riemann surface more from an algebraic way.
- The specific seminar plan can be seen by clicking this link

Senior Analysis Seminar - Riemannian Geometry Organizer: Bochen Liu — Guangdong, China, Sep. 2025 - Jan. 2026

- Reading the classical textbook *'Riemannian Geometry' - By Do Carmo* from chapters 0 to 7. Have a basic intuition for the metric structure on a manifold and how it helps us study its structure.

Geometric Models and Their additive Engineering Organizer: Yifei Zhu — Guangdong, China, Sep. 2025 - Jan. 2026

- Reading the general idea about the minimal surface and Pleatau's problem and

presenting a lecture.

- Printing some types of minimal surfaces, like the Catenoid, the Costa-Hoffman-Meeks Surface. Gain the basic ability to visualize the geometric model and print the 3D model to make research and communication much easier.
- The specific seminar plan can be seen by clicking this link

Mathematics Summer School

Summer School on Geometry and Topology, Sichuan University — Sichuan, China, July. 2026

- Attended three advanced mathematical lectures focusing on manifold topology, tiling theory, and noncommutative algebraic geometry.

Summer School on Differential Geometry, Xiamen University — Fujian, China, June. 2026

- Attended four advanced mathematical lectures and seminars focusing on differential geometry, mainly geometric measure theory and minimal submanifolds.

National Undergraduate Mathematics Competition Summer Camp — Shaanxi, China, Aug. 2025

- Attended advanced mathematical lectures and seminars from various fields.

Prizes and Awards

The Chinese Mathematics Competitions (16th & 17th), both First Award — Oct. 2024/Oct. 2025
CUMCM, Modeling Competition, Third Award/Second Award — Sept. 2024/Sept. 2025
Chinese Mathematics Olympics, Third Award — Sept. 2022
Chinese Physics Olympics, Third Award — Sept. 2021